

RM03

Number and Letter Series

About 3-5 marks in every RAS Prelims paper - series, analogy and odd-one-out taken together.

Confirmed appearances include 2013, 2016, 2018, 2021, 2023 and 2024. Difficulty is easy to moderate. These are the marks you bank first and come back to nothing.

This chapter is not knowledge, it is a checklist. A series question has a rule hidden in it, and there are only about a dozen rules RPSC ever uses. The student who runs the checklist finds the rule in forty seconds; the student who stares at the numbers hoping for inspiration loses two minutes and still guesses.

Series, Analogy, Classification

Toolkit (learn cold)

- A=1 to Z=26 positions
- EJOTY and CFILORUX anchors
- Reverse position = $27 - n$
- Squares, cubes, primes, Fibonacci

Wrong term

- Fix the rule from three agreeing terms
- Run it forward
- First term that breaks is the answer

Number series families

- Constant difference (AP)
- Differences in AP - n -squared type
- Constant ratio / rising multipliers
- prev $\times 2$ plus-or-minus k
- n^2+1 , n^2-1 , n^3+1 , n^3-1 , $n(n+1)$
- Primes, triangular, Fibonacci

Letter and alphanumeric

- Convert letters to numbers first
- Fixed gap, growing gap, squares as positions
- Opposite-letter pairs AZ BY CX DW
- Letter track and number track are independent

Two special shapes

- Alternate / interleaved - split odd and even places
- Alternating operations - $\times 3$ then -3

Analogy and odd one out

- Number: difference, ratio, square, cube, $n(n+1)$
- Letter: shift each letter, apply same shifts
- Word: name the category first
- Letter group: compare internal gaps

The alphabet table - memorise this before you touch a single

question

Start here. Almost every letter series, every letter analogy and every coding question in the paper is really a number question wearing a costume. You take the letters off the page, turn them into their positions in the alphabet, do arithmetic, and turn the answer back into a letter. If you have to count A-B-C-D on your fingers each time, you will lose thirty seconds an item and you will make counting errors. So learn the table below cold. It is the single highest-return thing in the whole reasoning section, because it pays in this chapter and again in Coding-Decoding.

Alphabet positions - learn both halves

Letter	No.	Letter	No.
A	1	N	14
B	2	O	15
C	3	P	16
D	4	Q	17
E	5	R	18
F	6	S	19
G	7	T	20
H	8	U	21
I	9	V	22
J	10	W	23
K	11	X	24
L	12	Y	25
M	13	Z	26

YAAD RAKHO

Remember EJOTY - E=5, J=10, O=15, T=20, Y=25. Five anchors at every multiple of five. Never count from A; jump to the nearest anchor and step two or three places. Second mnemonic for the in-between values: CFILORUX - C=3, F=6, I=9, L=12, O=15, R=18, U=21, X=24, the multiples of three. Between EJOTY and CFILORUX you are never more than one step from a known letter.

Reverse position and opposite letters - reverse position = 27 minus forward position

Letter	Forward position	Reverse position (27 - n)	Opposite letter
A	1	26	Z
B	2	25	Y
C	3	24	X
D	4	23	W
E	5	22	V
F	6	21	U
G	7	20	T
H	8	19	S
I	9	18	R
J	10	17	Q
K	11	16	P
L	12	15	O
M	13	14	N

TRAP

Shifts wrap round the alphabet. After Z comes A again, and before A comes Z. So A minus 3 is X, not a blank, and Z plus 2 is B. Students lose easy marks by stopping at the edge of the alphabet instead of going round it. Write the alphabet as a circle in your rough work if it helps.

The letter-to-number toolkit

EJOTY

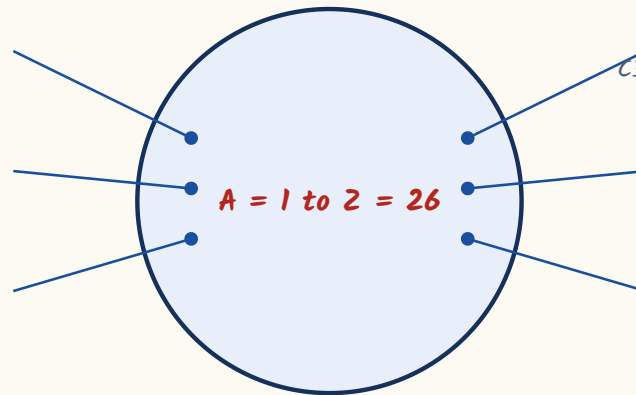
ES J10 O15 T20 Y25

Reverse position

27 minus forward position

Forward wrap

2 plus 2 gives B



CFILORUX

C3 F6 I9 L12 O15 R18 U21 X24

Opposite letters

A-Z, B-Y, C-X, M-N

Backward wrap

A minus 3 gives X

The attack order - what to test, and in what order

When a number series appears, do not think. Run the list. There is a fixed order because the cheap tests catch the common series and the expensive tests are only needed for the rare ones. Write the first differences under the series in your rough space every single time - that one habit solves more than half of all RAS series items on its own.

The six tests, always in this order

1. First differences - constant? then it is an AP, add the same again

2. Second differences - are the differences themselves in AP? then it is an n-squared type

3. Ratios - constant ratio is a GP; growing multipliers x2, x3, x4 also live here

4. Squares and cubes plus or minus a small constant - check terms against 1,4,9,16 and 1,8,27,64

5. Alternate terms - split into odd places and even places and solve each separately

6. Primes, triangular numbers, Fibonacci - the named lists

The families RPSC actually uses

Family	Rule	First terms
Squares	n^2	1, 4, 9, 16, 25, 36, 49
Cubes	n^3	1, 8, 27, 64, 125, 216
Square plus one	$n^2 + 1$	2, 5, 10, 17, 26, 37
Square minus one	$n^2 - 1$	0, 3, 8, 15, 24, 35
Cube plus one	$n^3 + 1$	2, 9, 28, 65, 126
Cube minus one	$n^3 - 1$	0, 7, 26, 63, 124, 215
Product of consecutives	$n(n+1)$	2, 6, 12, 20, 30, 42
Cube plus square	$n^3 + n^2$	2, 12, 36, 80, 150, 252
Triangular	differences 2,3,4,5	1, 3, 6, 10, 15, 21, 28
Fibonacci type	each term = sum of previous two	1, 1, 2, 3, 5, 8, 13, 21
Primes	no factor except 1 and itself	2, 3, 5, 7, 11, 13, 17, 19, 23

- Squares to know without thinking: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225.
- Cubes to know: 1, 8, 27, 64, 125, 216, 343, 512, 729, 1000, 1331, 1728.
- Primes under 60: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59.
- 1 is NOT prime. 2 is the only even prime - RPSC has used both of these as the trick.
- Constant difference means arithmetic; constant ratio means geometric; differences forming an AP means a quadratic, n-squared type.

YAAD RAKHO

Trick that saves the most time: if the differences themselves double (say 3, 6, 12, 24), the rule is almost never a difference rule at all - it is 'previous term times 2 plus or minus k'. Find k on one pair and confirm it on a second pair before you commit.

Differences and ratios – three worked examples

Work these on paper with me. The point is not the answer, it is the writing down of the differences. Take the series 5, 11, 19, 29, 41, ? - the numbers look irregular, but the differences are not.

- Series: 5, 11, 19, 29, 41, ?
- First differences: $11-5 = +6$, $19-11 = +8$, $29-19 = +10$, $41-29 = +12$.
- The differences are themselves rising by 2 - so the next difference is +14.
- Answer: $41 + 14 = 55$.
- Note what happened - test 1 failed (differences not constant) but test 2 caught it at once.

- Series: 6, 11, 21, 36, 56, ? - differences are +5, +10, +15, +20, an AP with common difference 5.
- Next difference is +25, so the answer is $56 + 25 = 81$.
- Series: 120, 99, 80, 63, 48, ? - differences are -21, -19, -17, -15, the odd numbers shrinking.
- Next difference is -13, so the answer is $48 - 13 = 35$.
- A falling series is the same drill. Do not panic at the minus signs; just write them down.

- Series: 4, 9, 19, 39, 79, ? - differences are +5, +10, +20, +40, which DOUBLE. Switch to the multiply rule.
- Test: $4 \times 2 + 1 = 9$. Confirm: $9 \times 2 + 1 = 19$, $19 \times 2 + 1 = 39$, $39 \times 2 + 1 = 79$. Rule holds.
- Answer: $79 \times 2 + 1 = 159$.
- Series: 7, 10, 16, 28, 52, ? - try previous $\times 2$ minus k. $7 \times 2 - 4 = 10$, and $10 \times 2 - 4 = 16$. Confirmed.
- Answer: $52 \times 2 - 4 = 100$.
- Series with rising multipliers: 3, 6, 18, 72, ? - the multipliers are $\times 2$, $\times 3$, $\times 4$, so next is $\times 5$ giving 360.
- Falling divisors work the same way: 625, 125, 25, 5, ? - each divided by 5, so the answer is 1.

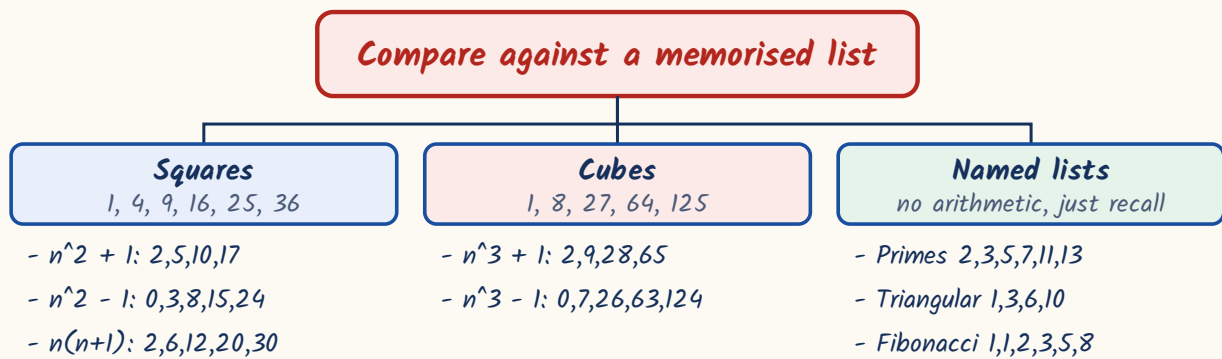
TRAP

Sometimes the OPERATION alternates rather than the number. Look at 5, 15, 12, 36, 33, 99, ? - nothing works as a difference or a ratio, because the steps are $\times 3$, then -3 , then $\times 3$, then -3 . Continue the pattern: $99 - 3 = 96$. Whenever alternate steps look sensible on their own but the whole series looks mad, suspect alternating operations.

Squares, cubes, primes and the offset families

This is test 4, and it is the one students skip. The moment a term sits one or two away from a square or a cube you know, stop doing differences and check the offset instead. Compare the series against the memorised list and write the offset above each term.

The offset families - known list, then the offset



- Series: 2, 5, 10, 17, 26, ? - compare with squares 1, 4, 9, 16, 25.
- Every term is exactly one more, so the rule is $n^2 + 1$.
- Answer: $6^2 + 1 = 37$. (Cross-check: the differences are the odd numbers 3, 5, 7, 9, 11 - that is the signature of a square-based series.)
- Series: 0, 7, 26, 63, 124, ? - compare with cubes 1, 8, 27, 64, 125. Every term is one less, so $n^3 - 1$.
- Answer: $6^3 - 1 = 216 - 1 = 215$.
- Same family shifted: 7, 26, 63, 124, 215, ? is $n^3 - 1$ starting from $n = 2$, so the answer is $7^3 - 1 = 342$.
- Harder mix: 2, 12, 36, 80, 150, ? - here $1+1$, $8+4$, $27+9$, $64+16$, $125+25$, so the rule is $n^3 + n^2$.
- Answer: $216 + 36 = 252$. When neither squares nor cubes fit alone, try their sum.

ASKED IN RAS

Asked in RAS Pre 2013 - complete 1, 4, 9, 16, 25, 36, ? Answer: 49. These are 1^2 to 6^2 , so the next is 7^2 . Note the distractor 48 sitting right next to it; RPSC always plants a near-miss.

ASKED IN RAS

Asked in RAS Pre 2021 - complete 2, 3, 5, 7, 11, 13, ? Answer: 17. These are consecutive primes. The trap option is 15, which looks like the right gap but is 3×5 and therefore composite.

Two more named lists finish this section. Triangular numbers 1, 3, 6, 10, 15, 21, 28 have differences 2, 3, 4, 5, 6 - so $21 + 7 = 28$. Fibonacci type means each term is the sum of the two before it: 1, 1, 2, 3, 5, 8, 13 gives $8 + 13 = 21$, and the same rule with different seeds gives 3, 4, 7, 11, 18, 29, 47. If a term looks like roughly the sum of the previous two, test it - it takes five seconds.

Alternate series, and finding the WRONG term

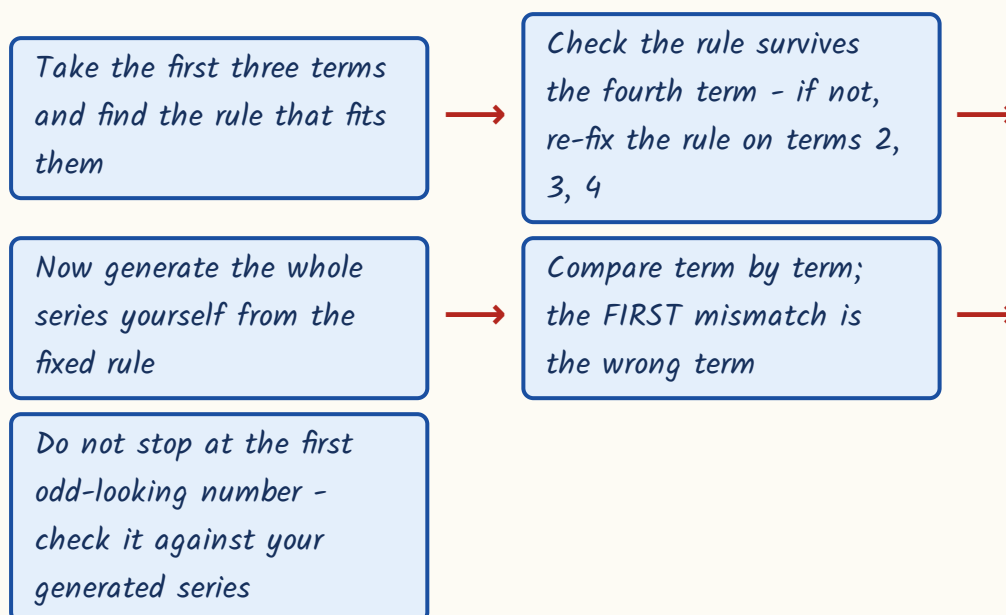
If no single rule fits the whole series, the series is probably two series plaited together. Split it: terms at the 1st, 3rd, 5th places form one series; terms at the 2nd, 4th, 6th places form another. Solve each separately and then answer for whichever place the question asks about. This is worth checking early whenever the terms zig-zag up and down.

- Series: 4, 9, 7, 18, 10, 27, 13, ? - the eighth term is asked, so it belongs to the even-place series.
- Odd places: 4, 7, 10, 13 - rising by 3.
- Even places: 9, 18, 27, ? - rising by 9.
- Answer: $27 + 9 = 36$.
- Series: 1, 2, 4, 6, 9, 12, 16, ? - odd places 1, 4, 9, 16 are perfect squares; even places 2, 6, 12, ? are $n(n+1)$.
- The eighth term is at an even place, so it is $4 \times 5 = 20$.

Missing term vs wrong term - two different jobs

Missing-term question	Wrong-term question
Every printed term obeys the rule	Exactly one printed term breaks the rule
Find the rule from any adjacent pair	Find the rule from THREE consecutive terms that agree
Extend the rule by one step	Run the rule from the start and find the first break
Answer is a new number not in the series	Answer is one of the numbers already printed
Options are close neighbours of the true value	Options are all terms of the given series

Wrong-term procedure



- Series: 3, 8, 15, 24, 34, 48, 63 - the first four are $4-1$, $9-1$, $16-1$, $25-1$, so the rule is $n^2 - 1$.
- Generated series: 3, 8, 15, 24, 35, 48, 63. The fifth printed term is 34 but should be 35. Answer: 34.
- Series: 2, 5, 10, 17, 26, 38, 50 - the rule is $n^2 + 1$, giving 2, 5, 10, 17, 26, 37, 50. Answer: 38.
- Series: 5, 10, 20, 40, 75, 160 - the rule is doubling, giving 5, 10, 20, 40, 80, 160. Answer: 75.
- Series: 1, 4, 9, 16, 24, 36, 49 - perfect squares; 24 should be 25. Answer: 24.

TRAP

In a wrong-term question the wrong number is usually only ONE away from the correct one - 34 for 35, 38 for 37, 24 for 25. That is deliberate. Never eyeball the series looking for the number that 'feels' out of place; generate the correct series in the margin and compare digit by digit.

Letter series and alphanumeric series

One rule governs the whole of this section: convert the letters into their positions, solve it as a number series, convert the answer back into a letter. Never guess by feel and never try to 'hear' the pattern. Write the number under each letter - it takes three seconds with the table you memorised and it removes the guesswork entirely.

- Series: A, C, F, J, O, ? - positions 1, 3, 6, 10, 15. Gaps: +2, +3, +4, +5.
- Next gap is +6, so position 21, which is U. Answer: U.
- Series: A, D, I, P, ? - positions 1, 4, 9, 16 are perfect squares. Next is 25 = Y. Answer: Y.
- Series: B, E, J, Q, ? - positions 2, 5, 10, 17, which is $n^2 + 1$. Next is 26 = Z. Answer: Z.
- Series: C, F, I, L, ? - constant gap of +3, so position 15 = O. Answer: O.
- Series: Z, X, V, T, ? - positions 26, 24, 22, 20 going down by 2. Next is 18 = R. Answer: R.

ASKED IN RAS

Asked in RAS Pre 2018 - complete A, C, F, J, O, ? Answer: U. The gaps grow by one each time (+2, +3, +4, +5, +6). If you had counted letters instead of writing positions you would very likely have landed on T.

ASKED IN RAS

Asked in RAS Pre 2023 - complete AZ, BY, CX, DW, ? Answer: EV. The first letters move forward A, B, C, D, E while the second letters move backward Z, Y, X, W, V. Notice these are exactly the opposite-letter pairs from the table in section 1 - each pair adds to 27.

Letter-group series - what to check

Series	What to look at	Answer
ACE, BDF, CEG, ?	Internal gap +2 fixed; each group starts one letter later	DFH
AB, DEF, HIJK, ?	Group length 2, 3, 4, 5; exactly one letter skipped between groups (C, G, L)	MNOPQ
AZ, BY, CX, DW, ?	First letter forward, second letter backward	EV
A, D, I, P, ?	Positions are perfect squares	Y

- **Alphanumeric series:** treat the letter track and the number track as two completely separate series.
- A1, C3, E5, G7, ? - letters skip one each time (A, C, E, G, I); the number is simply the letter's own position. Answer: I9.
- Z1, Y2, X3, W4, ? - letters run backwards, numbers run forwards. Answer: VS.
- A2, D4, G8, J16, ? - letters advance by 3 (A, D, G, J, M); numbers double (2, 4, 8, 16, 32). Answer: M32.
- 2A, 5D, 10G, 17J, ? - numbers are $n^2 + 1$ (2, 5, 10, 17, 26); letters advance by 3. Answer: 26M.
- Letter-number-SYMBOL series work the same way - a third track of symbols (*, @, #) usually repeats on a short cycle.
- Solve the easiest track first, then use it to knock out three options before you even check the others.

Analogy - number, letter and word

An analogy gives you a relation in the first pair and asks you to reproduce it in the second. Name the relation in words before you look at the options; if you read the options first you will find a relation that fits some option, which is not the same thing as the relation the question intends. For numbers test in this order - difference, ratio, square, cube, $n(n+1)$, n^2 plus or minus 1, digit reversal. For letters compute the positional shift for EVERY letter of the first pair and apply exactly those shifts. For words state the relation as a full sentence: 'a pen is used to write'. If two options both fit, your relation is too loose - make it more specific until only one survives.

- $8 : 27 :: 64 : ?$ - $8 = 2^3$ and $27 = 3^3$, cubes of consecutive numbers. $64 = 4^3$, so the answer is $5^3 = 125$.
- $25 : 36 :: 49 : ?$ - squares of consecutive numbers 5 and 6. $49 = 7^2$, so the answer is $8^2 = 64$.
- $6 : 42 :: 8 : ?$ - the relation is $n : n(n+1)$, since $6 \times 7 = 42$. So $8 \times 9 = 72$.
- In each case naming the relation ('cubes of consecutive numbers') is the whole solution; the arithmetic is trivial.
- $BDF : CEG :: MOQ : ?$ - every letter moves +1 (B to C, D to E, F to G). Apply +1 to M, O, Q: answer NPR.
- $ACE : BDF :: PRT : ?$ - again +1 on each letter. P to Q, R to S, T to U: answer QSU.
- Check every letter, not just the first. A rule that fits the first letter and fails the third is a wrong rule.
- If the shift is not constant, look for a shift that depends on the letter's place in the group (+1, +2, +3).

Word-analogy relations RPSC repeats

Relation	Example
Tool to its function	Pen : Write :: Knife : Cut
Specialist to organ treated	Ophthalmologist : Eye :: Cardiologist : Heart
Worker to workplace	Teacher : School :: Judge : Court
Product to raw material	Bread : Wheat :: Cloth : Cotton
Part to whole	Petal : Flower :: Chapter : Book
Animal to its young	Cow : Calf :: Horse : Foal
Cause to effect	Fire : Smoke :: Rain : Flood
Instrument to what it measures	Thermometer : Temperature :: Barometer : Pressure

ASKED IN RAS

Asked in RAS Pre 2024 - Ophthalmologist : Eye :: Cardiologist : ? Answer: Heart. Learn the whole set, because the distractors are drawn from it: nephrologist-kidney, neurologist-brain, pulmonologist-lung, dermatologist-skin, orthopaedist-bones.

Odd one out (classification), and how to play this block in the exam

Classification asks which item does not share the property the other three share. So the job is to name the property, not to find the strange item. Three of them will belong to one clean category. Find that category, and the fourth is the answer by elimination.

What to test, by type of item

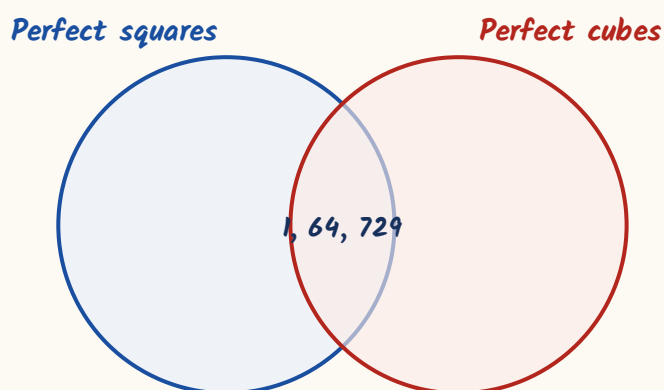
Type	Tests to run, in order
Numbers	Prime, perfect square, perfect cube, common factor, digit sum
Letter groups	Internal gaps between successive letters; the group with a different gap pattern is out
Words	Name the category - metals, flowers, alloys, birds that fly, capitals - then find the outsider
Mixed	Look for the property three share; never look for what makes one 'special'

- 100, 27, 64, 125 - 27, 64 and 125 are perfect cubes (3^3 , 4^3 , 5^3). 100 is a perfect square, not a cube. Answer: 100.
- 3, 9, 5, 11 - 3, 5 and 11 are prime; $9 = 3 \times 3$ is composite. Answer: 9.
- ACE, GIK, RTW, MOQ - gaps are +2,+2 in ACE, GIK and MOQ but +2,+3 in RTW. Answer: RTW.
- Rose, Mango, Lotus, Lily - three are flowers, mango is a fruit. Answer: Mango.
- Brass, Copper, Zinc, Iron - copper, zinc and iron are pure metals; brass is an alloy of copper and zinc. Answer: Brass.

ASKED IN RAS

Asked in RAS Pre 2016 - which is different: 100, 27, 64, 125? Answer: 100. Watch the built-in trap: 64 is both a square and a cube, so a careless student picks 64 as 'the special one'. The property shared by three of them is cube, and 64 has it.

Squares and cubes overlap - that is the bait



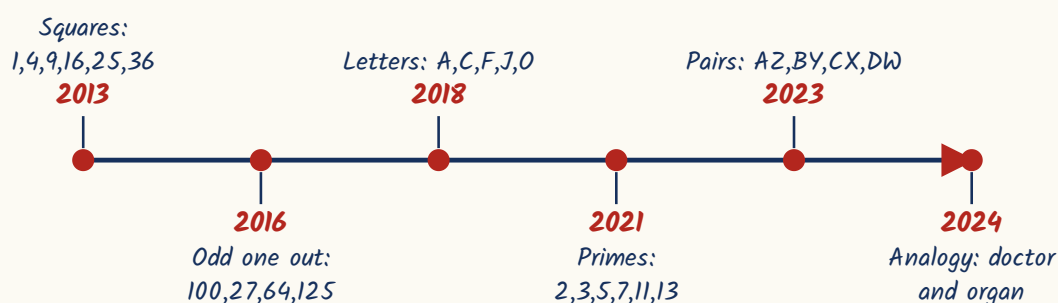
64 is both. In RAS Pre 2016 the answer was 100 - the only square among three cubes.

YAAD RAKHO

Remember for classification with numbers: if one option is a perfect square AND a perfect cube (1, 64, 729), it is almost never the answer - it is the bait. Ask what three of them share, never what one of them has extra.

Exam strategy for this block. Series, analogy and classification are the easiest scoring items in the RAS reasoning section, so do them on your first pass and bank the marks. But there is one-third negative marking, so a genuine blank is better than a wild guess. The practical rule: if you have found the rule and it fits every printed term, mark it and move on without re-checking. If ninety seconds have gone and no rule has appeared, leave it, and come back only if time remains at the end.

What RPSC actually asked in this block



REVISION RECAP

1. A=1 to Z=26 and reverse = 27 minus forward position. Memorise both; they pay in this chapter and in coding.
2. EJOTY: ES JIO OIS T20 Y25. CFILORUX: the multiples of 3. Jump to an anchor, never count from A.
3. Opposite pairs sum to 27: A-Z, B-Y, C-X, D-W, E-V, F-U, G-T, H-S, I-R, J-Q, K-P, L-O, M-N.
4. Shifts wrap: A - 3 = X, Z + 2 = B.
5. Attack order: first differences, second differences, ratios, squares/cubes with offset, alternate terms, primes.
6. Constant difference = AP; constant ratio = GP; differences in AP = n-squared type.
7. Differences that double mean 'previous x 2 plus or minus k' - fix k on one pair, confirm on a second.
8. Rising multipliers x2, x3, x4, x5 give 3, 6, 18, 72, 360. Falling divisors give 625, 125, 25, 5, 1.

9. Offset families: n^2+1 (2,5,10,17,26), n^2-1 (0,3,8,15,24), n^3-1 (0,7,26,63,124), $n(n+1)$ (2,6,12,20,30), n^3+n^2 (2,12,36,80,150).
10. Odd-number differences 3, 5, 7, 9 are the fingerprint of a square-based series.
11. If the series zig-zags, split it into odd places and even places and solve the two separately.
12. Alternating operations ($\times 3$, -3 , $\times 3$, -3) explain series that defy both differences and ratios.
13. Wrong term: fix the rule on three agreeing terms, generate the series yourself, take the FIRST mismatch.
14. The wrong term is planted only one away from the correct value - never eyeball it.
15. Letter series: convert to positions, solve as numbers, convert back. A,C,F,J,O gaps $+2,+3,+4,+5$ gives U.
16. Positions as squares: A, D, I, P, Y. Positions as n^2+1 : B, E, J, Q, Z.
17. Alphanumeric: letter track and number track are independent - solve the easier one and eliminate.
18. Number analogy order: difference, ratio, square, cube, $n(n+1)$, n^2 plus or minus 1, digit reversal.
19. Word analogy: say the relation in a full sentence before looking at the options.
20. Odd one out: name the property that THREE share. 64 as square-and-cube is bait, not the answer.